ON A QUESTION ABOUT WEAK POSITIVITY FOR PLURICANONICAL DIRECT IMAGES

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Abstract. We construct a pure Hodge module with globally generated lowest Hodge piece of rank two whose direct image after tensoring with the third relative canonical power is $\mathcal{O}_{\mathbf{P}^1}(-3)$, giving a negative answer to a question of Popa on weak positivity for pluricanonical direct images.

1. Introduction. Weak positivity of relative pluricanonical direct images is a theorem of Viehweg [12]: if $f : Z \to Y$ is a surjective morphism between smooth projective varieties, then $f_*\omega_{Z/Y}^{\otimes m}$ is weakly positive for every $m \geq 1$. Over a curve, these bundles are nef [1, Remark 1.7]. Popa–Schnell prove effective generation over higher-dimensional bases and recover weak positivity by vanishing methods [9]. On the other hand, lowest Hodge pieces of Hodge modules also satisfy positivity results under projective direct image [8, Theorems 47 and 55]. Popa asked whether these two positivity statements continue to hold together for higher pluricanonical powers.

For a Hodge module M on a smooth variety Z, let $E_Z(M)$ denote the lowest nonzero piece of its underlying filtered left $\mathscr{D}_Z$-module. When M is pure with strict support Z, its restriction to a dense open subset where M is given by a variation of Hodge structure is the highest nonzero Hodge bundle of that variation. Throughout, $E_Z(M)$ denotes the filtration piece defined on all of Z, including its extension across the boundary. In this notation, Popa's question takes the following form.

Question 1.1 ([8, Question 51])**.** Let $f : Z \to Y$ be a surjective morphism between smooth projective varieties, and let N be a mixed Hodge module on Z whose restriction to a nonempty open subset is a variation of mixed Hodge structure. Is

$$f_*\big(E_Z(N) \otimes \omega_{Z/Y}^{\otimes m}\big)$$

weakly positive for every $m \geq 1$?

For line-bundle twists, Păun–Takayama prove positivity under a semipositive singular metric and a fibrewise integrability condition [6]. The example below shows that the unrestricted statement fails even when the lowest Hodge piece is a globally generated vector bundle of rank two.

Put $X = \mathbf{P}^1 \times \mathbf{P}^1$, let $f : X \to \mathbf{P}^1$ be the second projection, and write $\mathcal{O}_X(a, b)$ for the product line bundles.

Main Theorem. *There is a rational polarizable pure Hodge module M with strict support X such that $E_X(M)$ is locally free and globally generated of rank two, while*

$$f_*\big(E_X(M) \otimes \omega_{X/\mathbf{P}^1}^{\otimes 3}\big) \simeq \mathcal{O}_{\mathbf{P}^1}(-3).$$

This work is supported by NSFC (Grant No. 12601116) and BJNSF (Grant No. 1264052).

More precisely, one can take

$$E_X(M) = E_0 \otimes \mathcal{O}_X(5,0),$$

where

$$0 \longrightarrow \mathcal{O}_X(-1,-3) \longrightarrow \mathcal{O}_X^{\oplus 3} \longrightarrow E_0 \longrightarrow 0. \tag{1.1}$$

The corresponding variation has rank two, type $(0,0)$*, and monodromy group* S_3*.*

Corollary 1.2. *Question 1.1 has a negative answer even for a pure Hodge module whose lowest Hodge piece is locally free and globally generated.*

Proof. Weak positivity of a vector bundle on a smooth projective curve is equivalent to nefness [2, Lemma 3.5]; see also [4, Proposition 6.2.11 and Theorem 6.1.10]. The direct image in the Main Theorem is the line bundle $\mathcal{O}_{\mathbf{P}^1}(-3)$, which is not nef. □

The bundle E_0 is constructed from three sections of $\mathcal{O}_X(1,3)$. A suitable twist determines a smooth finite flat triple cover through the construction of Miranda [5]; we use the formulation and smoothness criterion in [3, Theorem 3.2]. Finite duality and strictness of filtered direct image identify the trace-free summand with the lowest Hodge piece on all of X. The negative direct image is then obtained by twisting (1.1) and pushing it forward along f. Related positivity properties of vector bundles associated with finite covers were studied by Peternell–Sommese [7]; here the failure appears only after tensoring with a higher relative canonical power and taking another direct image.

The failure can also occur in all sufficiently large degrees for a fixed family and a fixed Hodge module. In Section 5, we construct a smooth product family with nonminimal general-type surface fibres and a fixed locally free, globally generated lowest Hodge piece of rank two for which weak positivity fails for every $m \geq 5$.

Acknowledgments. The author thanks Mihnea Popa for helpful correspondence and for clarifying the intended strict-support condition in Question 1.1.

2. Notations. Let Z be a smooth variety of dimension d. If $(\mathcal{M}, F_\bullet)$ is a filtered left $\mathscr{D}_Z$-module, the corresponding filtered right $\mathscr{D}_Z$-module is

$$\mathcal{M}^r = \omega_Z \otimes \mathcal{M}, \quad F_p\mathcal{M}^r = \omega_Z \otimes F_{p+d}\mathcal{M}.$$

Thus, if $E_Z(M)$ is the lowest nonzero piece of the filtered left $\mathscr{D}_Z$-module underlying a Hodge module M, and $S_Z(M)$ is the lowest nonzero piece in the right-module convention, then

$$S_Z(M) = \omega_Z \otimes E_Z(M).$$

We use the left-module notation $E_Z(M)$ in the statements and pass to right modules only when using filtered direct images.

For a morphism $f : Z \to Y$, we use f_* in two related senses. If $\mathcal{F}$ is a coherent sheaf on Z, then $f_*\mathcal{F}$ denotes its ordinary direct image. If M is a Hodge module, then f_*M denotes Saito's direct image in the derived category of Hodge modules. The underlying filtered $\mathscr{D}$-module is obtained from the filtered $\mathscr{D}$-module direct image. For projective f, this direct image is strict with respect to the Hodge filtration, so the filtration on each cohomology $\mathscr{D}$-module is computed at each fixed filtration index before taking cohomology. For the finite morphisms $g : Y \to X$ used below, the direct image is concentrated in degree zero. If $\dim X = \dim Y = d$, the

lowest right Hodge piece of $g_*\mathbf{Q}_Y^H[d]$ is $g_*\omega_Y$; see [8, (13)]. This identification is compatible with the trace map.

3. The bundle E_0. All varieties are over $\mathbf{C}$. We keep X and f as above and let $q = \mathrm{pr}_1$ be the first projection. For a vector bundle E, we use the convention $\mathbf{P}_X(E) = \mathrm{Proj}_X \operatorname{Sym} E$, so $\pi_*\mathcal{O}_{\mathbf{P}(E)}(a) = \operatorname{Sym}^a E$ for $a \geq 0$. The following lemma gives the bundle in (1.1).

Lemma 3.1. *There are three sections generating $\mathcal{O}_X(1,3)$ at every point such that the quotient E_0 in (1.1) satisfies*

$$E_0|_{q^{-1}(x)} \simeq \mathcal{O}_{\mathbf{P}^1}(1) \oplus \mathcal{O}_{\mathbf{P}^1}(2), \quad E_0|_{f^{-1}(t)} \simeq \mathcal{O}_{\mathbf{P}^1} \oplus \mathcal{O}_{\mathbf{P}^1}(1)$$

for every $x, t \in \mathbf{P}^1$. Moreover, for $V_0 = \operatorname{Sym}^3 E_0 \otimes \det(E_0)^{-1}$, one has

$$q_*V_0 \simeq \mathcal{O}_{\mathbf{P}^1}(-4)^{\oplus 10}, \quad R^1q_*V_0 = 0.$$

The bundle $V_0 \otimes \mathcal{O}_X(k,0)$ is globally generated for every $k \geq 4$.

Proof. The triples of sections of $\mathcal{O}_X(1,3)$ form an irreducible parameter space. At each point of X, a common zero imposes three independent linear conditions. Since X is a surface, the incidence of such zeros has image of positive codimension in the parameter space. A general triple therefore generates $\mathcal{O}_X(1,3)$ at every point. Regard the coefficients in the second factor as a pencil of 3×4 matrices parametrized by the first factor $\mathbf{P}^1$. Matrices of rank at most two have codimension two in $\mathbf{P}^{11}$, so a general projective line avoids them. Thus the three cubics on every fibre of q are linearly independent for a general triple. Both conditions define nonempty open subsets of the same parameter space.

The quotient E_0 is locally free and globally generated. On a fibre of q it has degree three. A trivial quotient would give, by dualizing (1.1) on that fibre, a constant linear relation among the three cubics. Both summands consequently have positive degree, so the splitting is $\mathcal{O}(1) \oplus \mathcal{O}(2)$. On a fibre of f, global generation and degree one give the splitting $\mathcal{O} \oplus \mathcal{O}(1)$.

It follows that V_0 restricts to $\mathcal{O} \oplus \mathcal{O}(1) \oplus \mathcal{O}(2) \oplus \mathcal{O}(3)$ on every fibre of q. Cohomology and base change show that $B = q_*V_0$ is a bundle of rank ten, that $R^1q_*V_0 = 0$, and that $q^*B \to V_0$ is surjective. The symmetric cube of the quotient in (1.1), twisted by $\det(E_0)^{-1} = \mathcal{O}_X(-1,-3)$ gives

$$0 \longrightarrow \mathcal{O}_X(-2,-6)^{\oplus 6} \longrightarrow \mathcal{O}_X(-1,-3)^{\oplus 10} \longrightarrow V_0 \longrightarrow 0. \tag{3.1}$$

This is the degree-three part of the symmetric algebra modulo the single linear relation in (1.1). Pushing forward by q gives

$$0 \longrightarrow B \longrightarrow \mathcal{O}_{\mathbf{P}^1_x}(-2)^{\oplus 30} \longrightarrow \mathcal{O}_{\mathbf{P}^1_x}(-1)^{\oplus 20} \longrightarrow 0.$$

Hence $\deg B = -40$. We show that the sections can be chosen so that $H^0(B(3)) = 0$.

Choose homogeneous coordinates x_0, x_1 on the first factor and t_0, t_1 on the second, and write

$$s_i = \sum_{h=0}^{1} \sum_{j=0}^{3} c_{ihj} x_0^{1-h} x_1^h t_0^{3-j} t_1^j, \quad 0 \leq i \leq 2.$$

Twisting (3.1) by $\mathcal{O}_X(3,0)$ identifies $H^0(B(3))$ with the kernel of

$$\mathbf{C}^6 \otimes H^0(\mathbf{P}^1, \mathcal{O}(1)) \otimes H^1(\mathbf{P}^1, \mathcal{O}(-6)) \longrightarrow \mathbf{C}^{10} \otimes H^0(\mathbf{P}^1, \mathcal{O}(2)) \otimes H^1(\mathbf{P}^1, \mathcal{O}(-3)),$$

where the H^0 and H^1 factors refer to the first and second factors of X, respectively. Both sides have dimension 60.

We describe this map in monomial bases. Index the columns by (u, j, b), where $u \in \mathbf{Z}^3_{\geq 0}$, $|u| = 2$, $0 \leq j \leq 4$, and $0 \leq b \leq 1$, and the rows by (v, a, d), where $|v| = 3$, $0 \leq a \leq 1$, and $0 \leq d \leq 2$. If e_i is the ith standard basis vector, then

$$\mathsf{M}_{(v,a,d),(u,j,b)} = \sum_{i=0}^{2} \mathbf{1}_{v=u+e_i}\, c_{i,d-b,j-a},$$

where coefficients with indices outside their prescribed ranges are zero. The multi-indices record multiplication in the symmetric powers, while the remaining indices come from multiplication in the first factor and, by Serre duality, the transpose of multiplication of a linear form by a cubic in the second factor.

Order the degree-e monomials as $(a, b, e-a-b)$, first by increasing a and then by increasing b, and order the remaining indices as above. For

$$(c_{0hj}) = \begin{pmatrix} 0 & 2 & -1 & -1 \\ 0 & 2 & 0 & 2 \end{pmatrix}, \quad (c_{1hj}) = \begin{pmatrix} -3 & -1 & 1 & 0 \\ -1 & 2 & -3 & 2 \end{pmatrix},$$

$$(c_{2hj}) = \begin{pmatrix} 3 & 1 & -2 & -3 \\ -2 & 3 & 2 & 3 \end{pmatrix},$$

Gaussian elimination modulo 101 gives

$$\det \mathsf{M} \equiv 41 \pmod{101}.$$

Hence the determinant polynomial in the coefficients c_{ihj} is nonzero. Since the parameter space is irreducible, its nonvanishing locus meets the open subset of triples satisfying the preceding conditions.

Choose a triple in this intersection. Since $H^0(B(3)) = 0$, every line summand of B has degree at most -4. As B has rank ten and degree -40, it follows that

$$B \simeq \mathcal{O}_{\mathbf{P}^1}(-4)^{\oplus 10}.$$

The surjection $q^*B \to V_0$ then shows that $V_0 \otimes \mathcal{O}_X(k, 0)$ is globally generated for every $k \geq 4$. □

Fix E_0 as in Lemma 3.1, and put $E_k = E_0 \otimes \mathcal{O}_X(k, 0)$. We use $k = 5$ for the Main Theorem and $k = 4$ in Section 5. The triple-cover construction of Miranda and Casnati–Ekedahl expresses a cover as a cubic divisor in $\mathbf{P}_X(E_k)$. We follow [3, Theorem 3.2 and equation (9)] and include the duality calculation identifying its Hodge coefficient.

4. Proof of the Main Theorem. For $k \geq 4$, set $\pi : P = \mathbf{P}_X(E_k) \to X$ and $L = \mathcal{O}_P(3) \otimes \pi^* \det(E_k)^{-1}$. Lemma 3.1 shows that

$$\pi_* L = \operatorname{Sym}^3 E_k \otimes \det(E_k)^{-1} = V_0 \otimes \mathcal{O}_X(k, 0)$$

is globally generated. The relative evaluation map then shows that L is globally generated. By Bertini, a general section cuts out a smooth divisor $Y \subset P$. A general section of the rank-four bundle $\pi_* L$ is everywhere nonzero on the surface X, since vanishing at a fixed point imposes four independent linear conditions. Hence Y meets each fibre of π in a subscheme of length three. The relative Cartier divisor construction therefore gives a finite flat map $g = \pi|_Y : Y \to X$ of degree three.

The canonical formula for a projective bundle and adjunction give

$$\omega_{P/X} = \mathcal{O}_P(-2) \otimes \pi^* \det(E_k), \quad \omega_{Y/X} = \mathcal{O}_P(1)|_Y.$$

Push forward the exact sequence $0 \to \omega_{P/X} \to \mathcal{O}_P(1) \to \mathcal{O}_Y(1) \to 0$. Relative duality gives $\pi_*\omega_{P/X} = 0$ and $R^1\pi_*\omega_{P/X} = \mathcal{O}_X$, whereas $\pi_*\mathcal{O}_P(1) = E_k$ and $R^1\pi_*\mathcal{O}_P(1) = 0$. We obtain $0 \to E_k \to g_*\omega_{Y/X} \to \mathcal{O}_X \to 0$. The last map is the trace on dualizing sheaves. Finite duality and the trace splitting in characteristic zero give

$$g_*\mathcal{O}_Y \simeq \mathcal{O}_X \oplus E_k^\vee, \quad g_*\omega_{Y/X} \simeq \mathcal{O}_X \oplus E_k. \tag{4.1}$$

On a fibre of f, the dual bundle restricts to $\mathcal{O}(-k) \oplus \mathcal{O}(-k-1)$, so $H^0(X, E_k^\vee) = 0$. Consequently $H^0(Y, \mathcal{O}_Y) = \mathbf{C}$, and Y is connected. Since it is smooth, it is integral.

On the étale locus $U \subset X$ of g, take the trace-free subspace of the rational permutation local system on the three sheets. It is a rank-two variation of type $(0,0)$, polarized by the restriction of the usual positive definite form on $\mathbf{Q}^3$. Let M_k be its pure extension with strict support X, given by Saito's correspondence [11, Theorem 3.21 and Section 4.1]. A finite map is small, so the finite direct image $g_*\mathbf{Q}_Y^H[2]$ is the intermediate extension of the permutation variation on U. Its constant and trace-free parts have strict support X. Strictness of projective filtered direct image [10] identifies their initial right Hodge pieces with the corresponding summands of $g_*\omega_Y$. Both initial indices are -2, because the variations have type $(0,0)$ on a surface. Using (4.1), we obtain

$$S_X(M_k) = \ker\bigl(g_*\omega_Y \longrightarrow \omega_X\bigr) = \omega_X \otimes E_k.$$

Thus $E_X(M_k) = E_k$ on all of X, including the discriminant. In particular, the coefficient is locally free and globally generated.

To determine the monodromy, we first show that E_0 is indecomposable. Suppose that $E_0 = L_1 \oplus L_2$ for line bundles L_1, L_2. The splittings in Lemma 3.1 force their degrees on the two factors to be $\{0,1\}$ and $\{1,2\}$, respectively. Hence

$$\int_X c_2(E_0) = \int_X c_1(L_1)c_1(L_2)$$

would be either 1 or 2. On the other hand, if

$$H_1 = c_1\mathcal{O}_X(1,0), \qquad H_2 = c_1\mathcal{O}_X(0,1),$$

then (1.1) gives

$$c_1(E_0) = H_1 + 3H_2, \qquad c_2(E_0) = 6H_1H_2,$$

and therefore $\int_X c_2(E_0) = 6$, a contradiction. Thus E_0 is indecomposable, and so is $E_k = E_0 \otimes \mathcal{O}_X(k,0)$. Since the triple cover is connected, its monodromy group is a transitive subgroup of S_3, hence either C_3 or S_3. If it were C_3, the complex trace-free local system would split as the sum of the two nontrivial characters. Their intermediate extensions and lowest Hodge pieces would then give a decomposition of E_k into two line bundles, contradicting its indecomposability. Therefore the monodromy group is S_3, and the trace-free variation is irreducible over $\mathbf{C}$.

Take $k = 5$. Since $\omega_{X/\mathbf{P}^1} = \mathcal{O}_X(-2,0)$, we have $E_X(M_5) \otimes \omega_{X/\mathbf{P}^1}^{\otimes 3} = E_0(-1,0)$. Twisting (1.1) gives

$$0 \longrightarrow \mathcal{O}_X(-2,-3) \longrightarrow \mathcal{O}_X(-1,0)^{\oplus 3} \longrightarrow E_0(-1,0) \longrightarrow 0.$$

The middle term has vanishing zeroth and first direct images, while

$$R^1 f_* \mathcal{O}_X(-2,-3) = \mathcal{O}(-3).$$

Hence $f_* E_0(-1,0) = \mathcal{O}(-3)$. Taking $M = M_5$ proves the theorem.

Remark 4.1. *Let $g : Y \to X$ be one of the triple covers constructed in the proof above, with coefficient E_k. If R_g is its ramification divisor, then $\omega_{Y/\mathbf{P}^1} = g^*\omega_{X/\mathbf{P}^1} \otimes \mathcal{O}_Y(R_g)$. Equation (4.1) and projection formula give*

$$(fg)_*\big(\omega_{Y/\mathbf{P}^1}^{\otimes m}(-(m-1)R_g)\big) \simeq f_*\omega_{X/\mathbf{P}^1}^{\otimes m} \oplus f_*\big(E_k \otimes \omega_{X/\mathbf{P}^1}^{\otimes m}\big).$$

For $m = 1$, the ramification term disappears. For $m > 1$, the coefficient direct image comes from the proper subsheaf $\omega_{Y/\mathbf{P}^1}^{\otimes m}(-(m-1)R_g)$ of the pluricanonical bundle. This explains how its negative summand can coexist with scalar pluricanonical weak positivity.

5. Higher degrees. For the fixed-coefficient construction, we need the other twists of (1.1). The pushforward calculation above and the fibre splitting in Lemma 3.1 give

$$f_* E_0(\ell,0) = \begin{cases} \text{a globally generated bundle}, & \ell \geq 0, \\ \mathcal{O}_{\mathbf{P}^1}(-3), & \ell = -1, \\ 0, & \ell \leq -2. \end{cases} \tag{5.1}$$

Indeed, for $\ell \geq 0$ pushing forward the twisted sequence gives $0 \to \mathcal{O}(-3)^{\oplus \ell} \to \mathcal{O}^{\oplus 3(\ell+1)} \to f_* E_0(\ell,0) \to 0$. For $\ell \leq -2$, its restriction to every fibre of f is $\mathcal{O}(\ell) \oplus \mathcal{O}(\ell+1)$ and its space of sections is zero.

Let B_1, B_2 be smooth projective curves of genus two, put $S_0 = B_1 \times B_2$, and write $K_0 = K_{S_0}$. The line bundle $\mathcal{O}_{S_0}(3K_0)$ is very ample, since $3K_{B_i}$ has degree six and is very ample on each factor. A general pencil in $|3K_0|$ has a transverse base locus of length $(3K_0)^2 = 72$, because $K_0^2 = 8$. Blow up these points, writing $\sigma : S \to S_0$ and R for the sum of the exceptional curves. The pencil gives a morphism $q_S : S \to \mathbf{P}^1$. Thus S is a smooth nonminimal surface of general type.

Lemma 5.1. *For every $m \geq 5$, the bundle $Q_m = q_{S*}\omega_S^{\otimes m}$ on $\mathbf{P}^1$ has exactly 72 copies of $\mathcal{O}_{\mathbf{P}^1}(-5)$ in its splitting into line bundles.*

Proof. The direct image Q_m is torsion-free on the smooth curve $\mathbf{P}^1$, hence is locally free. The pencil and blowup formulas give

$$q_S^*\mathcal{O}(1) = \sigma^*\mathcal{O}_{S_0}(3K_0) \otimes \mathcal{O}_S(-R), \quad \omega_S = \sigma^*\mathcal{O}_{S_0}(K_0) \otimes \mathcal{O}_S(R).$$

For $0 \leq j \leq m$ it follows that

$$\omega_S^{\otimes m} \otimes q_S^*\mathcal{O}(j) = \sigma^*\mathcal{O}_{S_0}((m+3j)K_0) \otimes \mathcal{O}_S((m-j)R).$$

For every integer $a \geq 0$, one has $\sigma_*\mathcal{O}_S(aR) = \mathcal{O}_{S_0}$. A meromorphic function with poles allowed only along the exceptional curves extends across their images, which have codimension two in the normal surface S_0. Projection formula therefore gives

$$h^0(\mathbf{P}^1, Q_m(j)) = h^0(S_0, (m+3j)K_0) = (2m+6j-1)^2 \quad (0 \leq j \leq m).$$

The last equality follows from Riemann–Roch on the two genus-two factors:

$$h^0(B_i, uK_{B_i}) = 2u - 1 \quad (u \geq 2).$$

Write $h(j) = h^0(Q_m(j))$. A summand $\mathcal{O}(a)$ contributes $\max\{a+j+1, 0\}$ to $h(j)$. Its contribution to $h(5) - 2h(4) + h(3)$ is one if $a = -5$ and zero otherwise. Since $m \geq 5$, the formula applies at $j = 3, 4, 5$, and its second difference is 72. □

Theorem 5.2. *There exist a smooth nonminimal surface S of general type and a rational polarizable pure Hodge module N with strict support $Z = S \times \mathbf{P}^1$ such that $E_Z(N)$ is locally free and globally generated of rank two. Its variation has rank two and type $(0,0)$. For the second projection $f_Z : Z \to \mathbf{P}^1$, one has*

$$f_{Z*}\big(E_Z(N) \otimes \omega_{Z/\mathbf{P}^1}^{\otimes m}\big) \simeq \mathcal{O}_{\mathbf{P}^1}(-3)^{\oplus 72} \oplus P_m \quad (m \geq 5),$$

where P_m is globally generated. The surface, family, and Hodge module are fixed independently of m.

Proof. Choose the smooth connected triple cover $g : Y \to X$ constructed above with coefficient E_4, and let $h = qg : Y \to \mathbf{P}^1$. The maps h and q_S have finitely many critical values. After composing q_S with an automorphism of $\mathbf{P}^1$, we may assume that these finite sets are disjoint. The fibre product $Y' = S \times_{\mathbf{P}^1} Y$ is then smooth: at every point at least one of dq_S and dh is surjective, so their difference is surjective.

Put $Z = S \times \mathbf{P}^1$, and let $v = q_S \times \mathrm{id} : Z \to X$. The finite projection $g' : Y' \to Z$ is the flat base change of g. Its algebra and dualizing module satisfy

$$g'_*\mathcal{O}_{Y'} \simeq \mathcal{O}_Z \oplus (v^*E_4)^\vee, \quad g'_*\omega_{Y'/Z} \simeq \mathcal{O}_Z \oplus v^*E_4.$$

On each $\mathbf{P}^1$ over a point of S, the dual coefficient restricts to $\mathcal{O}(-1) \oplus \mathcal{O}(-2)$ by Lemma 3.1. Its space of global sections is therefore zero, so $H^0(Y', \mathcal{O}_{Y'}) = \mathbf{C}$. Thus Y' is connected.

Take the trace-free part of the permutation variation of g' and its pure extension N to Z. As in the proof of the Main Theorem, smallness of the finite map gives strict support Z. Strict filtered direct image and finite duality identify

$$S_Z(N) = \ker\big(g'_*\omega_{Y'} \longrightarrow \omega_Z\big) = \omega_Z \otimes v^*E_4.$$

Here the variation has type $(0,0)$ on the threefold Z, so its initial right index is -3. Thus $E_Z(N) = v^*E_4$, a locally free and globally generated bundle of rank two. The family and coefficient are now fixed.

The relative canonical bundle of the product family is $\omega_{Z/\mathbf{P}^1} = \mathrm{pr}_S^*\,\omega_S$. Flat base change along the second factor gives $v_*\,\mathrm{pr}_S^*\,\omega_S^{\otimes m} = Q_m \boxtimes \mathcal{O}_{\mathbf{P}^1}$. Projection formula and composition of direct images then identify

$$f_{Z*}\big(E_Z(N) \otimes \omega_{Z/\mathbf{P}^1}^{\otimes m}\big) \simeq f_*\big(E_0(4,0) \otimes (Q_m \boxtimes \mathcal{O}_{\mathbf{P}^1})\big).$$

A line summand $\mathcal{O}(a)$ of Q_m contributes $f_*E_0(4+a, 0)$. By (5.1), this contribution is $\mathcal{O}(-3)$ when $a = -5$ and is globally generated or zero otherwise. Lemma 5.1 gives exactly 72 copies of $\mathcal{O}(-5)$ for every $m \geq 5$. Their contributions give the stated decomposition, and the direct sum of the remaining contributions is the globally generated bundle P_m. □

Since $f_Z : Z = S \times \mathbf{P}^1 \to \mathbf{P}^1$ is the product projection, its ordinary pluricanonical direct images are $H^0(S, \omega_S^{\otimes m}) \otimes \mathcal{O}_{\mathbf{P}^1}$. The blowup $\sigma : S \to S_0$ preserves plurigenera,

so Riemann–Roch on the two genus-two factors of S_0 gives

$$f_{Z*}\omega_{Z/\mathbf{P}^1}^{\otimes m} \simeq \begin{cases} \mathcal{O}_{\mathbf{P}^1}^{\oplus 4}, & m = 1, \\ \mathcal{O}_{\mathbf{P}^1}^{\oplus (2m-1)^2}, & m \geq 2. \end{cases}$$

Thus these direct images are nonzero trivial bundles in every positive degree.

In contrast, the direct images in Theorem 5.2 have a quotient $\mathcal{O}_{\mathbf{P}^1}(-3)$ for every $m \geq 5$ and are therefore not weakly positive, by the same argument as in Corollary 1.2. Each exceptional curve $E \subset S$ satisfies $K_S \cdot E = -1$, and their contribution to the section counts in Lemma 5.1 produces the negative summands.

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